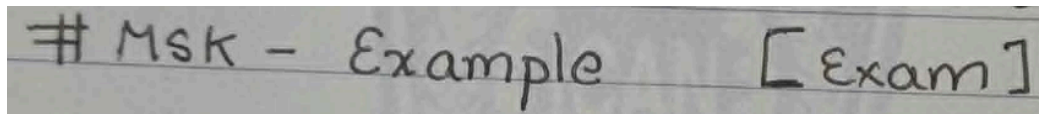


Day 9 Of Communication System Class

• Problem w/ QAM (PSK)
- phase change $\rightarrow$ distortion
- smooth transition भ्रम

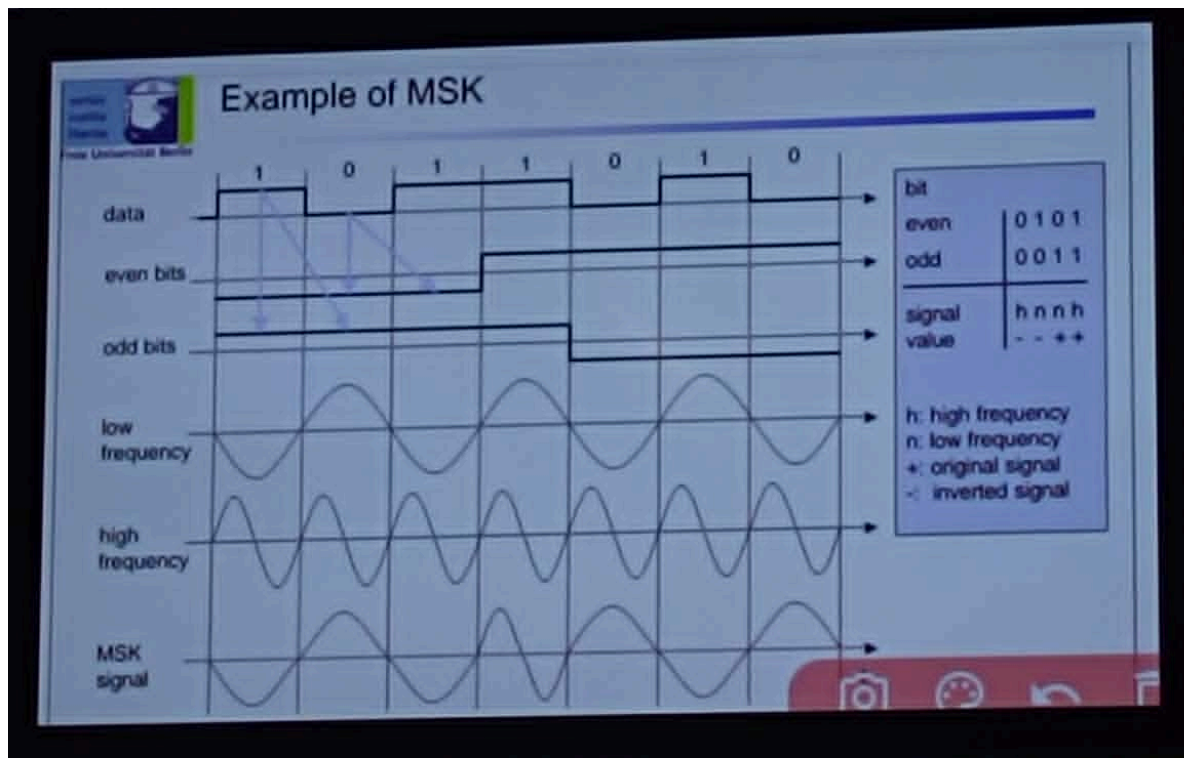
[MSK - Multiple SK
QSK - Gaussian SK] आर्के
पढ़ने
☐: slides

[Most Probable Questions]

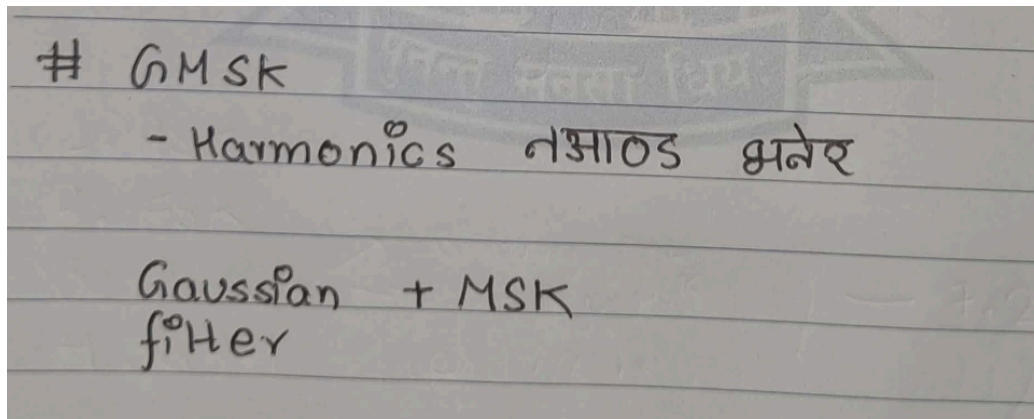
Minimum Shift Keying (MSK)

MSK - A special case of binary continuous phase FSK modulation technique in which the change in carrier frequency from symbol 0 to symbol 1 or vice versa is exactly equal to one-half the bit rate of input data signal.

$$v_{\text{MSK}}(t) = \begin{cases} \sqrt{\frac{2E_b}{T_b}} \cos [2\pi f_1 t + \theta(0)] & \text{for binary 1} \\ \sqrt{\frac{2E_b}{T_b}} \cos [2\pi f_2 t + \theta(0)] & \text{for binary 0} \end{cases}$$

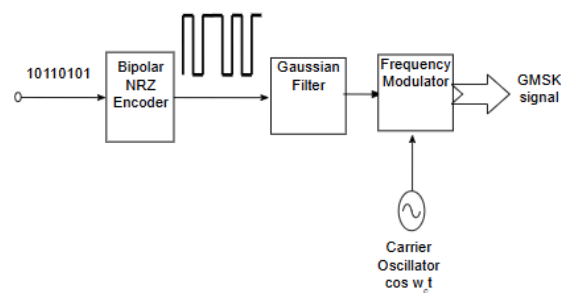


• Problem w/ Minimum SK
- Bandwidth Management

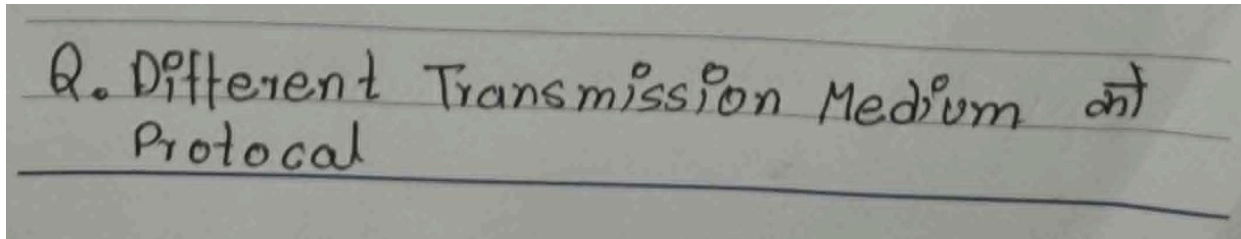


Gaussian Minimum Shift Keying (GMSK)

GMSK - A special case of MSK in which a pre-modulation low-pass Gaussian filter is used as a pulse shaping filter to reduce the bandwidth of the baseband signal before it is applied to MSK modulator.



[Most Probable Questions]



Q. Write About Different Transmission Mediums With Their Protocols .

Transmission Medium	Core Type	Primary Protocol(s)	Typical Use Case
Twisted Pair	Wired	Ethernet (802.3), PPPoE	Office LANs, DSL Internet
Coaxial Cable	Wired	DOCSIS	Cable Internet, TV Networks
Fiber Optic	Wired	GPON, SONET/SDH, Fibre Channel	High-speed Internet Backbones, FTTH
Radio Waves	Wireless	Wi-Fi (802.11), Bluetooth, 5G	Mobile networks, Smart home devices
Microwaves	Wireless	DVB-S2, WiMAX	Satellite communication, Long-range towers
Infrared	Wireless	IrDA	Remote controls, short-range peripherals